

Sunbeam Platform

Post-U.S. Global Infrastructure Solutions

Sunbeam Studios

The Three Problems

Availability

Data flows through U.S. infrastructure you don't control, subject to laws not written with your interests in mind.

Vulnerability

U.S.-headquartered players have legal access to your data held by U.S. companies anywhere in the world.

Affordability

Over \$20,000/yr for required network security you need to pay for by U.S. companies.

**The United States controls
70% of the global internet**

Why Now?

- European Parliament voted 471–68 to reduce reliance on foreign technology
- NIS2 in force — must demonstrate verifiably effective security
- Cyber Resilience Act phases in through 2027
- Sovereign cloud spending projected to triple to \$23 billion by 2027

Team



**Sienna Meridian
Satterwhite**

**Studio Head & Chief
Engineer**



Lonneke Faber

Head of Production



Tony McKenzie

Head of Engineering

Sunbeam Proxy

Sunbeam Proxy

The security proxy that proves itself.

Most tools work from a list of known threats. If an attack isn't on the list, it gets through.

Sunbeam Proxy learns what normal traffic looks like for your server and flags anything that doesn't fit — stopping threats no rulebook has seen before.

Proxy architecture diagram

Every other security product says
“trust us, it works.”

Sunbeam Proxy proves that
it does.

How It Works

1. **DDoS Detection** — machine learning monitors IP behaviour to spot flooding, abnormal rates, or coordinated attacks
2. **Scanner Detection** — machine learning examines requests for automated probing or access to vulnerable paths
3. **Cloud-Native** — integrates natively through Kubernetes Gateway API or Envoy xDS

Each layer uses a compact ML model under 4 KB, deciding in under 6 nanoseconds.

Four Guarantees

Liveness

The models never fail to return a verdict.

Stability

Attackers cannot game the models.

Hardware-faithful

Account for processor rounding and numeric behaviour.

Verifiable

No adversarial perturbation can change the verdict.

The Market

EU sovereign infrastructure:
\$23 billion by 2027.

No credible open-source
Cloudflare alternative for
Europe. Incumbents are US-
controlled, closed-source, or
lack formal verification.

That structural gap creates a
durable opening.

Provider	Monthly Cost
Sunbeam Proxy	€1,250
AWS Shield Advanced	€2,800
Azure DDoS Protection	€2,750

The Studio's Self-Hosted Stack

WFE

Persistent, embeddable workflow engine for Rust. Saga compensation, event-driven pausing, distributed locking. Code or YAML definitions.

Sol

Multiplayer AI agent for Matrix. Group chats with team context, DMs with per-user memory, dedicated coding agent. Self-hosted. E2E encrypted.

CesiumDB

Rust-native LSM-tree store. 646,000 writes/second. Bloom filter lookups at 860 picoseconds. Namespaces, HLC versioning, background compaction.

Business Model

All products are open source. Core software is free to deploy indefinitely.

Sunbeam Proxy free tier = the full product:

- Full ML threat detection
- Formal verification
- CROWN certification
- Up to 1 Gbps at no cost

Paid offerings target scale, managed operations, support, and compliance.

Three Revenue Streams

Self-hosted Licensing

Same binary, higher
bandwidth caps.

Gross margin: 88–
92%

Managed Service

Sunbeam operates on
EU-sovereign
infrastructure via
Scaleway co-location.

Gross margin: 65–70%

Enterprise Support

Formal verification
artefacts, CROWN
methodology,
deployment
assistance, NIS2 and
Cyber Resilience Act
compliance materials.

Financial Projections

Phase	ARR Target	Year	ARR
Phase 1	€9K–15K	Year 1	€1M–€2M
Phase 2	€66K–123K	Year 2	€3M–€5M
Phase 3	€180K–300K	Year 3	€8M–€15M

Key Milestones:

- Phase 1: Open-source launch, CNCF Sandbox, training data resolved
- Phase 2: First Enterprise customers, managed beta, EU grants
- Phase 3: Series A ready, 400–800 Gbps benchmark, 100+ deployments

Commercialisation

Sunbeam Proxy is our commercial product.

WFE and CesiumDB are open source forever — free to use, no commercial layer planned.

Sol is open source today. Commercial licensing will be formalised as the product matures.

Sunbeam Studios aspires to
create beauty, to create
community, and to insoure
the world

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